center of a black hole, but not for the critical circumference. This image made it difficult for physicists to accept the Oppenheimer–Snyder conclusion that a person who rides through the Schwarzschild singularity (the critical circumference) on an imploding star will feel *no* infinite gravity and see *no* breakdown of physical law.

How truly *nonsingular* the Schwarzschild singularity (critical circumference) is did not become fully clear until David Finkelstein discovered his new reference frame and used it to show that the Schwarzschild singularity is nothing but a location into which things can fall but out of which nothing can come—and a location, therefore, into which we on the outside can never see. An imploding star continues to exist after it sinks through the Schwarzschild singularity, Finkelstein's reference frame showed, just as the Sun continues to exist after it sinks below the horizon on Earth. But just as we, sitting on Earth, cannot see the Sun beyond our horizon, so observers far from an imploding star cannot see the star after it implodes through the Schwarzschild singularity. This analogy motivated Wolfgang Rindler, a physicist at Cornell University in the 1950s, to give the Schwarzschild singularity (critical circumference) a new name, a name that has since stuck: He called it the *horizon.*

There remained the issue of what to call the object created by the stellar implosion. From 1958 to 1968 different names were used in East and West: Soviet physicists used a name that emphasized a distant astronomer's vision of the implosion. Recall that because of the enormous difficulty light has escaping gravity's grip, as seen from afar the implosion seems to take forever; the star's surface seems never quite to reach the critical circumference, and the horizon never quite forms. It looks to astronomers (or would if their telescopes were powerful enough to see the imploding star) as though the star becomes frozen just outside the critical circumference. For this reason, Soviet physicists called the object produced by implosion a *frozen star*—and this name helped set the tone and frame of mind for their implosion research in the 1960s.

In the West, by contrast, the emphasis was on the viewpoint of the person who rides inward on the imploding star's surface, through the horizon and into the true singularity; and, accordingly, the object thereby created was called a *collapsed star.* This name helped focus physicists' minds on the issue that became of greatest concern to John Wheeler: the